Question 6: Implement Kubernetes deployment strategies such as rolling updates and rollbacks, and demonstrate scaling of the application replicas.

A rolling update from image 2.0.0 to 2.0.1 was performed and observed, the change was rolled back, and the deployment was scaled from 2 to 4 replicas and back.

Rolling update

```
$ docker build --build-arg APP_VERSION=2.0.1 -f docker/Dockerfile -t devops-lab2-app:2.0.1 .
$ minikube image load devops-lab2-app:2.0.1
$ kubectl set image deployment/lab2-app lab2-app=devops-lab2-app:2.0.1
deployment.apps/lab2-app image updated

pod transitions captured during the rollout:
-- t=5  1/1 Running (new)   1/1 Running (new)   Terminating (old)
-- t=8  1/1 Running (new)   1/1 Running (new)   Terminating (old) x2

$ curl http://192.168.49.2:30080/health
{"status":"UP","version":"2.0.1"}
```

Rollback

```
$ kubectl rollout undo deployment/lab2-app
deployment.apps/lab2-app rolled back
$ kubectl get pods -l app=lab2-app -o jsonpath image field
lab2-app-88f4c9fc9-bv4qd:  devops-lab2-app:2.0.0
lab2-app-88f4c9fc9-snrr5:  devops-lab2-app:2.0.0

$ curl http://192.168.49.2:30080/health   (after old pods terminated)
{"status":"UP","version":"2.0.0"}
```

Scaling

```
$ kubectl scale deployment/lab2-app --replicas=4
deployment.apps/lab2-app scaled

$ kubectl get pods -l app=lab2-app
4 pods Running
$ kubectl get deployment lab2-app
lab2-app  4/4  4  4

$ kubectl scale deployment/lab2-app --replicas=2
$ kubectl get pods -l app=lab2-app
2 pods Running
```

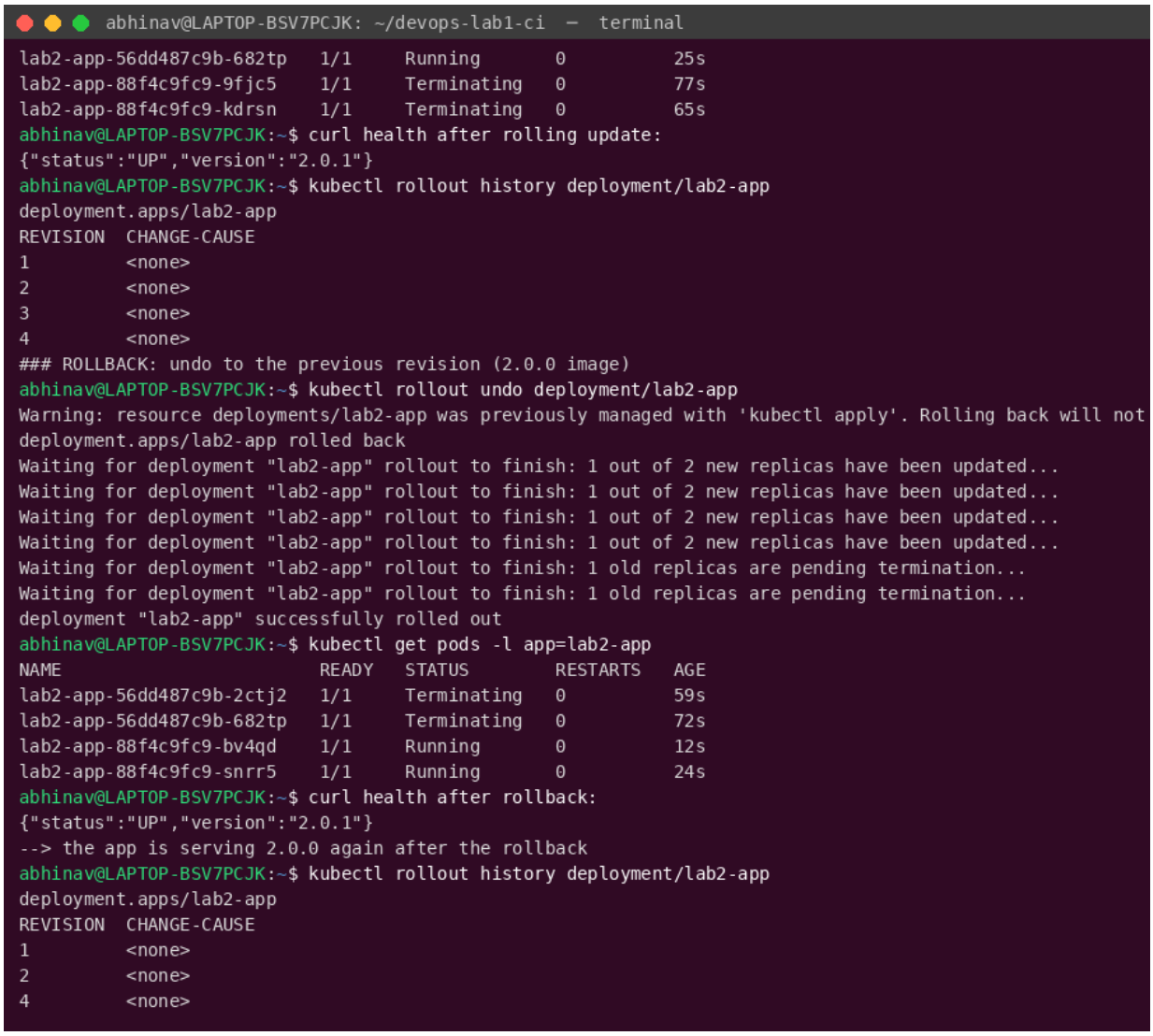


Fig: Rolling update: new pods Running while old pods terminate

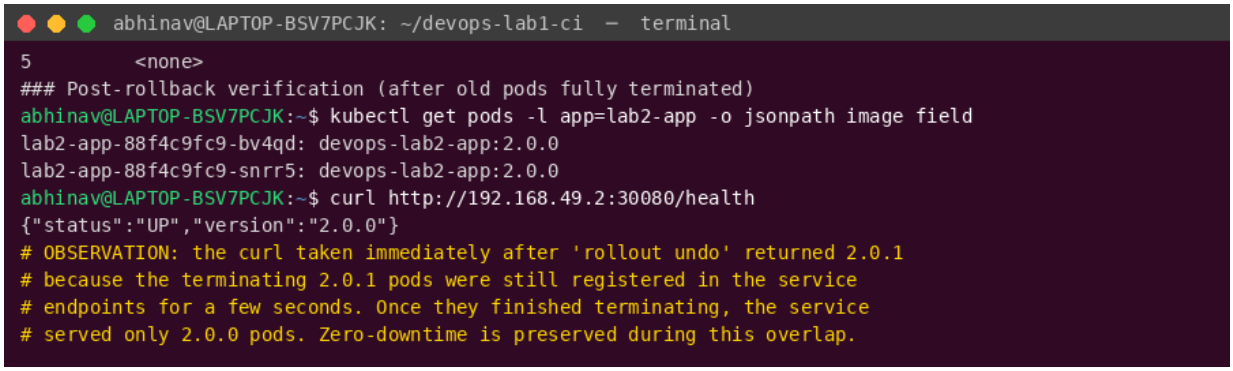


Fig: Rollback verified: 2.0.0 image running again

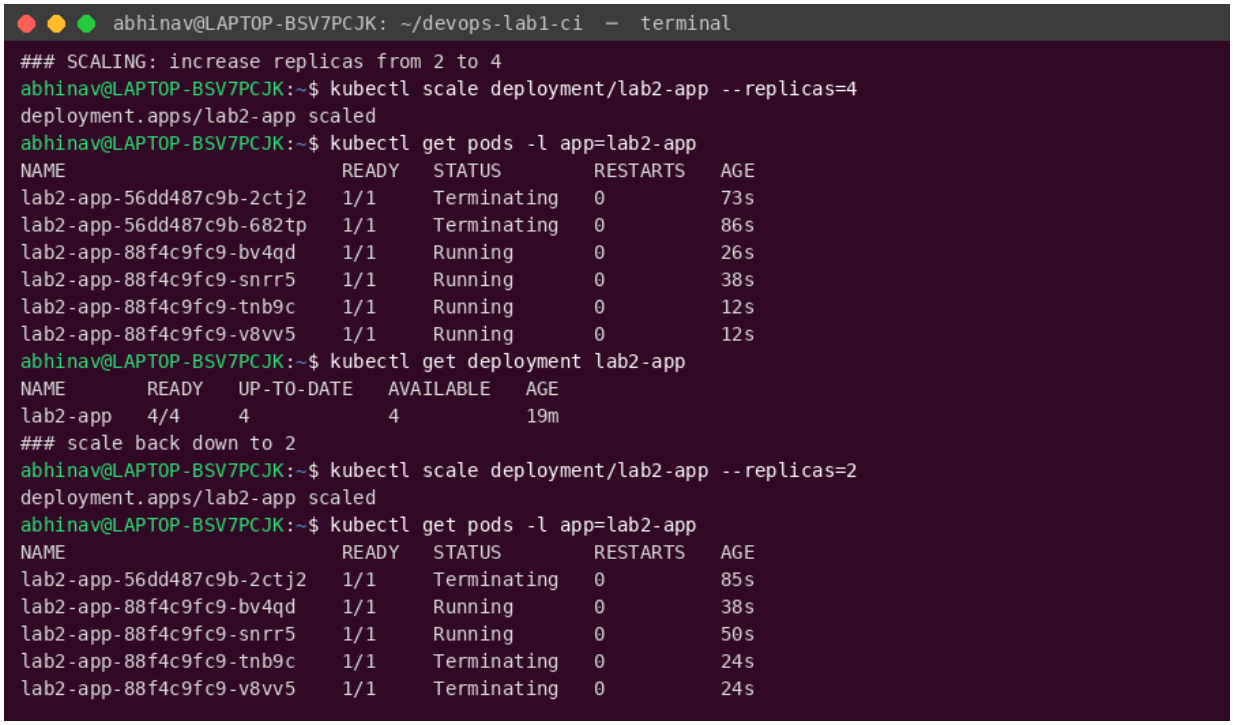


Fig: Scaling 2 -> 4 -> 2 replicas

Observations

- The rolling update replaced pods one by one - the service never went down (captured transitions show new pods Running before old pods finish Terminating).
- The version reverts with the image because APP_VERSION is baked in at build time (build ARG) rather than set through the ConfigMap.
- The health curl taken immediately after `rollout undo` still returned 2.0.1 for a few seconds: the terminating pods were still registered in the service endpoints.
- Scaling is instantaneous on a single-node cluster - two extra Running pods appeared within seconds.

Conclusion

All experiments were performed on Ubuntu 24.04 LTS with Docker, Docker Compose, Minikube and kubectl. The evidence (command logs, screenshots and commit history) is committed in the GitHub repository github.com/theabhinav0231/devops-lab-2 under the docs/ directory.